The empirical recurrence for OEIS A268788: recovering a conjecture that is not written down

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Abstract

OEIS A268788 records "Empirical recurrence of order 68", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 177$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A268788 is "Number of nX7 binary arrays with some element plus some horizontally, vertically or antidiagonally adjacent neighbor totalling two exactly once.". It has offset 1 and begins

$$38, 850, 14984, 228384, 3295246, 45515227, 611932378, 8057509992, \ \ldots$$

The entry states:

> Empirical recurrence of order 68 (see link above)

As of the "Last modified" line on the live entry (Feb 13 2016 12:26:07, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 177$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 177$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 354$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{68} c_i\, a(n-i),$$

c_1	2	c_{18}	−40352987546	c_{35}	−2516449266318	c_{52}	392403435
c_2	169	c_{19}	−1731681754	c_{36}	−5299006856467	c_{53}	121804600
c_3	694	c_{20}	169766303246	c_{37}	4612288755692	c_{54}	−56942553
c_4	−7705	c_{21}	−172083944226	c_{38}	−370411564193	c_{55}	−45190096
c_5	−81246	c_{22}	−432283697779	c_{39}	−1670643254910	c_{56}	−2736961
c_6	−222034	c_{23}	1174345160584	c_{40}	881083365721	c_{57}	8047064
c_7	517502	c_{24}	−213727795318	c_{41}	254923161724	c_{58}	3525933
c_8	4048112	c_{25}	−3410566119732	c_{42}	−375390897947	c_{59}	53154
c_9	1950100	c_{26}	6248797331497	c_{43}	18817875590	c_{60}	−513311
c_{10}	−34597333	c_{27}	−1297199940236	c_{44}	107541892314	c_{61}	−229598
c_{11}	−49390138	c_{28}	−14252531123312	c_{45}	−21340705994	c_{62}	−42389
c_{12}	222391082	c_{29}	31972773053846	c_{46}	−26802863398	c_{63}	1728
c_{13}	411980164	c_{30}	−36845671545670	c_{47}	6427885660	c_{64}	2624
c_{14}	−1316829914	c_{31}	22145471845712	c_{48}	6733400845	c_{65}	486
c_{15}	−2225642966	c_{32}	1754955549210	c_{49}	−1016537708	c_{66}	3
c_{16}	7627596454	c_{33}	−16611707402840	c_{50}	−1726204662	c_{67}	−10
c_{17}	7520225330	c_{34}	14194215422343	c_{51}	−78736390	c_{68}	−1

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68+4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

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